ITA06020 MACHINE LEARNING

Experiment 19 – Bank Loan Prediction Using Naive Bayes

Aim:
To predict credit approval risk using the Naive Bayes classification algorithm.

Procedure:

1. Import the required Python libraries.
2. Load the German Credit dataset.
3. Convert categorical data into numerical values.
4. Split the dataset into training and testing data.
5. Train the Gaussian Naive Bayes model.
6. Predict the credit risk and calculate accuracy.

Dataset: German Credit Dataset (real-world credit risk dataset available through Scikit-learn's OpenML interface). It contains information about applicants and their credit history.

Python Code:

```
import pandas as pd

from sklearn.datasets import fetch_openml

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix

credit = fetch_openml(name="credit-g", version=1, as_frame=True)

X = credit.data

y = credit.target

X = pd.get_dummies(X, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42, stratify=y

)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")

sample = X_test.iloc[[0]]
```

```
prediction = model.predict(sample)

print("Predicted Credit Risk:", prediction[0])

print("Actual Credit Risk:", y_test.iloc[0])

print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred))
```

Output:

```
Accuracy: 65.5 %
Predicted Credit Risk: good
Actual Credit Risk: good
Confusion Matrix:
[[36 24]
 [45 95]]
```

Result:

The Naive Bayes algorithm was successfully implemented to classify applicants into good or bad credit risk categories.